

# RAGHAVENDRA BASAVARAJ

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## SUMMARY AND OBJECTIVE

- Graduate course concentration in RF/Microwave design passionate to pursue a career in the RF and wireless industry.
- Hands on practical experience with RF lab equipments like **VNA, spectrum analyzer, signal generator**, and **oscilloscopes**.
- Competent in DSP, RFIC | RF switches, RF system design, SAW/BAW filters, MMIC design, and in designing advanced RF transceiver system.

## EDUCATION

### Masters of Science in Electrical Engineering

(Jan 2017 – Dec 2018)

University of South Florida, Tampa, FL, USA

GPA: 3.83/4.0

### Bachelor of Engineering in Electronics & Communication

(Sep 2011 – June 2015)

Visvesvaraya Technological University, India

GPA: 78.3% (3.5/4.0)

## SKILLS

- **CAD Tools:** Keysight ADS, Ansys HFSS, Matlab and MathCad.
- **RF Hardware:** Experienced in configuring, de-embedding and calibrating the test setup in the design and measurement of Advanced transceiver architectures, Mixers, Lumped and Distributed Low Pass Filters, Couplers, Amplifiers, Patch Antennas, Waveguide Filters and non-linear measurements such as ACPR, P1dB compression point and third order intercept (OIP3).
- **Analog Circuit design:** Two stage Opamp design, Current mirrors, LDO, Band gap references, CMRR, PSRR etc.
- **Antennas:** Antenna fundamentals, Antenna measurements, TRP, TIS and SAR measurements, design of dipole and monopole antennas, loop and slot antennas, horn antennas, broadband micro strip antennas, and Antenna arrays.
- **Communications:** RF Link Budget Analysis, Link Margin, Desensitization measurements, Receiver dynamic range measurements, Digital and Analog Modulation Schemes, Error Control and Coding.
- **Signal Processing:** FIR and IIR filters, Nyquist sampling and aliasing techniques, Channel Capacity, Coherent Receiver design.
- **Wireless Standards:** Familiar with Multiple Access protocols like CDMA, TDMA, OFDM, OFDMA, LTE, GPS, Cryptography.
- **Certifications:** Keysight RF/MW Industry-Ready Certified Student | Graduate Certificate in Wireless Engineering from the University of South Florida, Tampa, FL, USA. | ITIL V3 Foundation Course - ITIL/2000308007 | CompTIA Server+ Certification - COMP001020903667 | RF Fundamentals, Basic Concepts and Components – RAHRF101

## RELEVANT COURSEWORK

- **Radio Frequency:** RF and microwave active and passive circuit design, RF Measurements, Analog Circuit Design, Wireless systems and communication laboratory, Power Amplifier design, Antenna theory, Monolithic microwave integrated circuit (MMIC) design, and Semiconductor device theory.
- **Communications:** Wireless network architecture, Digital communication systems, and Digital Signal Processing.

## WORK EXPERIENCE

### Independent Study, *University of South Florida*

(Dec 2017 – Nov 2018)

- The idea was to build an X-band **band pass filter** inside a rectangular waveguide by the use of resonators to achieve high linearity. The design was simulated using HFSS. **WR-90** was made use at a cut off frequency of **6.56GHz** with **TE<sub>101</sub>** mode of operation.

### Technology Consultant in Storage, *Hewlett-Packard Enterprise*

(Sep 2015 – Dec 2016)

- Experienced in troubleshooting various issues in HPE Storage Devices such as MSA, 3PAR, XP, EVA, Brocade Switches etc.
- Experienced working in Mission Critical Environments as a '**Technical Account Manager**'.

## ACADEMIC PROJECTS

### Design of a RF/Microwave low Noise Amplifier:

The transistor, "BFP 420 NPN BJT" was utilized for the design of the Low Noise Amplifier at **1.4GHz**. Conjugate matching was employed and ultimately an amplifier with a low noise of about **1.47dB** and high gain of **15.75dB** was designed, fabricated and tested using ADS.

### Design of a Travelling wave (Distributed Amplifier) using the supplied OMMIC (D01MH) design kit:

This project aims in designing a distributed broadband amplifier with a gain of **12+/-1dB** from **dc to 40GHz** in the mm wave frequency range with an equivalent Noise figure to be less than **4.0dB**, (i/p and o/p **VSWR<2**). The design includes cascaded connection of HEMT transistors along with inductors forming artificial transmission line. Non-linear measurements (**P1dB>15dBm** and **IIP3**) are performed and verified.

### Design of RF Power Amplifier:

The design of a class AB power amplifier using Qorvo GaN HEMT was performed. The amplifier had to be designed for the following specifications: Center frequency=**1.8 GHz**, Gain = **16dB**, PAE = **60%**. Transistor was stabilized from **DC-3 GHz**. Design and simulation was carried out using ADS. Output power of amplifier was **49 dBm**, with a **gain of 16.7dB** and **PAE of 60.14%**.

**Mini Projects:** Design and analysis of **coupled line Band pass filter** at **2.1GHz** | Design and implementation of a **Micro strip Impedance Matching circuit** in ADS | **Bias circuit design and experiment** | Design and analysis of **microstrip quadrature coupler** in ADS |